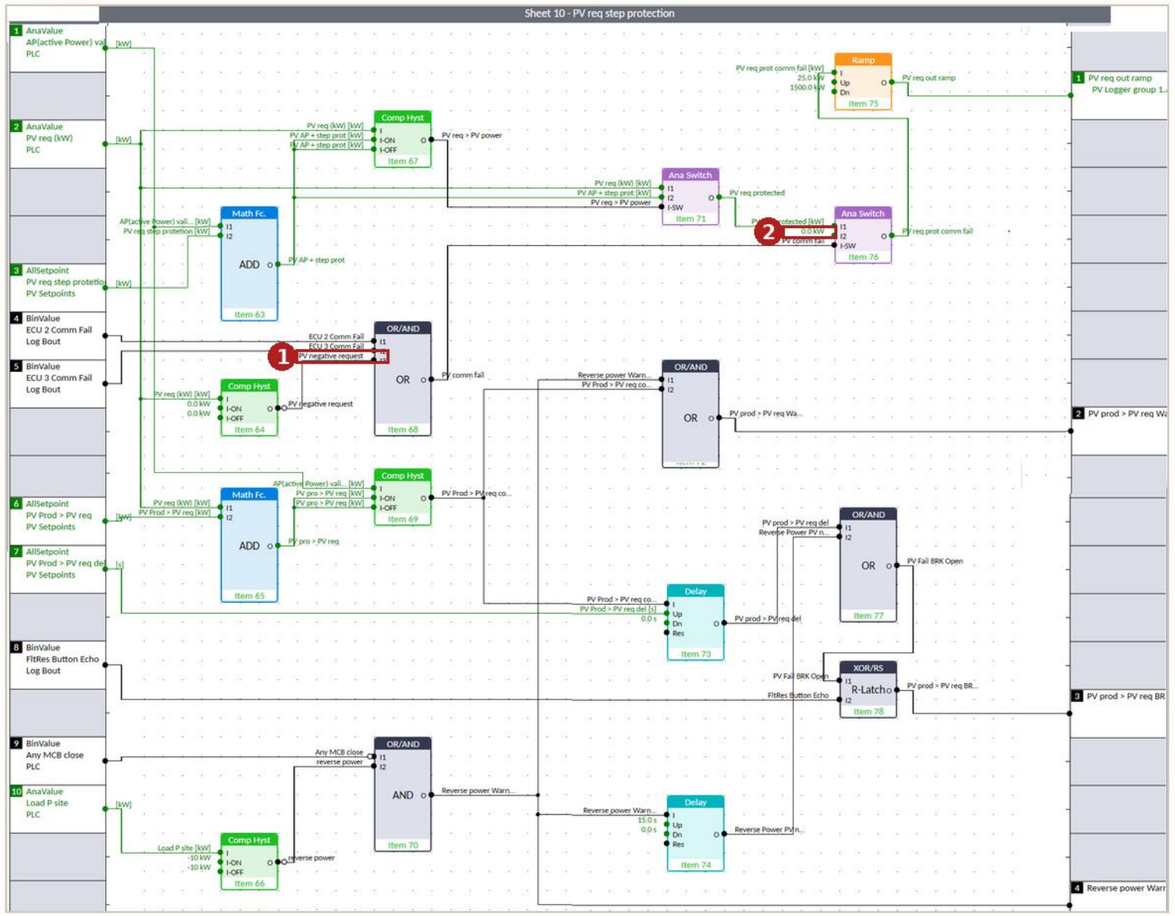
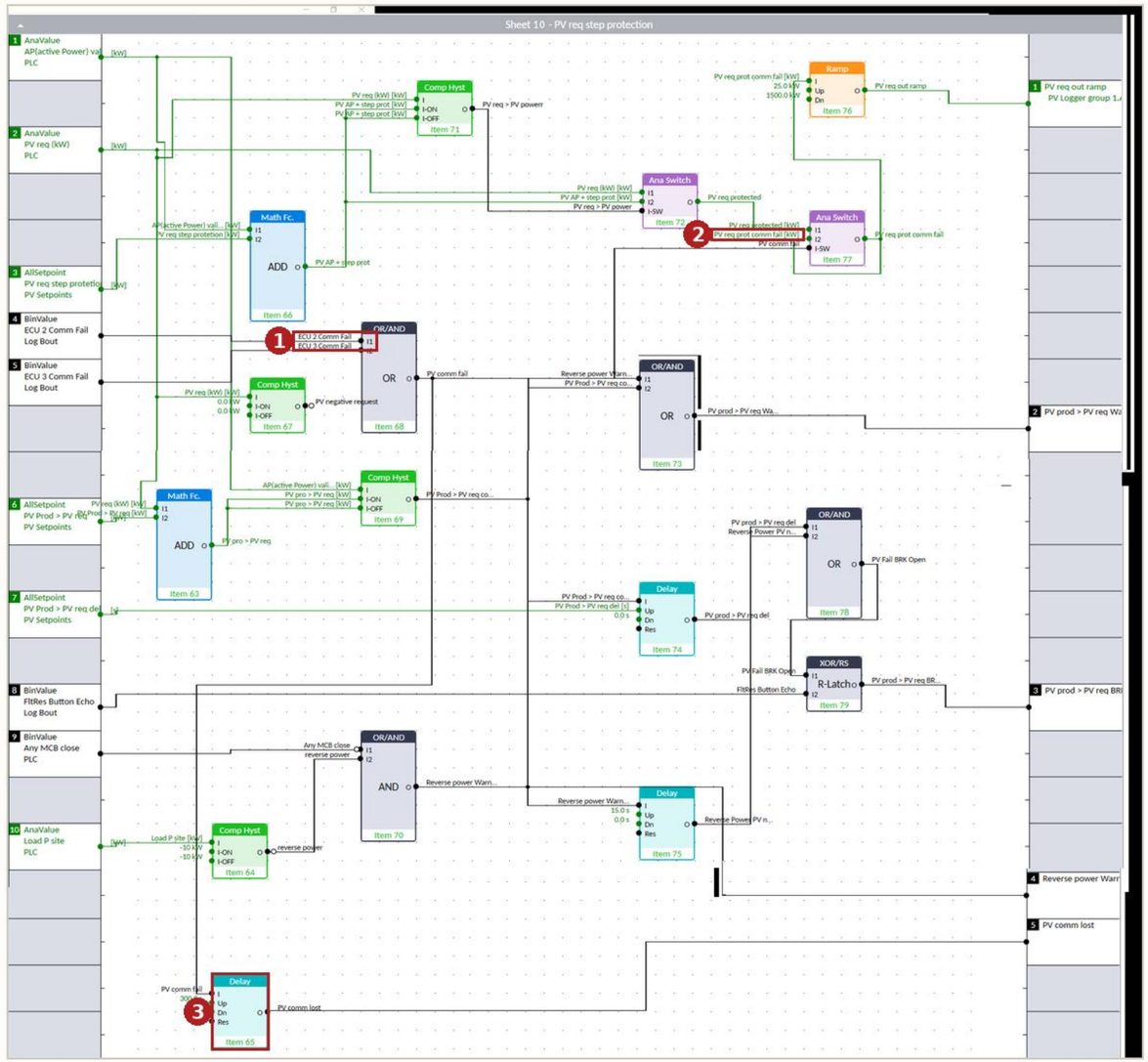
The PV logger comm fail: what the controller does today, and the small change that stops it

Today



Item 76 is a switch. When either PV logger session misses one reply, Item 68 flips it to its second position, a fixed 0.0 kW (2). That zero goes to the SAU as the solar setpoint. The inverters obey within seconds, the gensets take the whole plant, the UPS are shed, and the solar needs about five minutes to climb back, 100 kW at a time. Five to thirty times a day. Item 68 (1) also fires on a request of zero or below.

After the change



The same switch, now Item 77: its second input is its own output (2), so a missed reply just repeats the last setpoint. Nothing drops. Item 68 reacts only to the two comm fail signals (1). One new timer, Item 65 (3), raises the flag PV comm lost after five minutes of silence, for an alarm. Nothing else on the sheet changed.

Why it matters, the risk, and why it should go in

Why it matters

Every comm fail is a production dip and a load step on the gensets. The history of 14 October shows three bursts of them, dozens of warnings, and the UPS shed at the worst moment. The zero the program sends does not even reach the logger while the link is down; it arrives when the link is back, when there is nothing left to protect. We pay every time for a protection that cannot act.

The risk

While the link is down the inverters keep their last setpoint, exactly as they do today; the change does not alter that. What could go wrong in a blind window is a large load drop with the solar still at its setpoint, pushing the gensets towards reverse power. The gensets hold 25 percent of their nominal in hand (788 kW with three running, 290 kW with one), the load of this plant moves a couple of hundred kW in half an hour, the controller's own protections do not depend on the logger, and after five minutes the flag is raised. The risk of leaving it as it is, we see every day.

Why it should go in

It is one wire, one input and one timer. With a healthy link the program behaves exactly as today, checked block by block against the plant's own history. It is undone in a minute: load the 14 October archive, or put the 0.0 kW back on the switch. And it is the precondition for the step that saves the fuel: running fewer gensets is only safe once a lost reply no longer takes the solar down.

In numbers

Model of 14 October with the change: three of the eleven UPS sheddings gone, about 350 kWh of solar back, 3.5 percent less fuel. With power management on afterwards: 30 percent less fuel and more.